

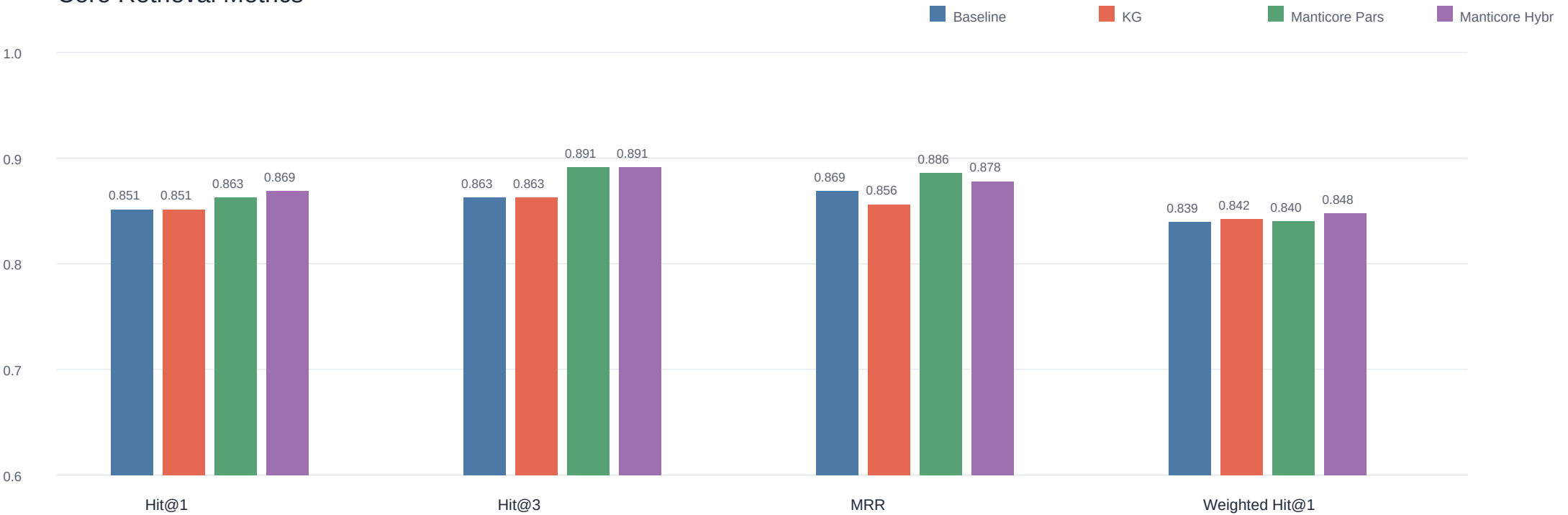
RTL Retrieval Methods: Performance Comparison

Dataset: 175 multi-axis questions, 732 indexed modules

Source: <https://github.com/manticoresoftware/manticoresearch> | generated 2026-05-09T20:02:34

Method	Hit@1	Hit@3	MRR	Weighted Hit@1	Avg query ms
Baseline	0.8514	0.8629	0.8686	0.8393	28.171
KG	0.8514	0.8629	0.8562	0.8423	34.284
Manticore Parser/LSP	0.8629	0.8914	0.8857	0.8401	37.386
Manticore Hybrid	0.8686	0.8914	0.8781	0.8479	46.418

Core Retrieval Metrics



Method Characteristics and Difficulty Breakdown

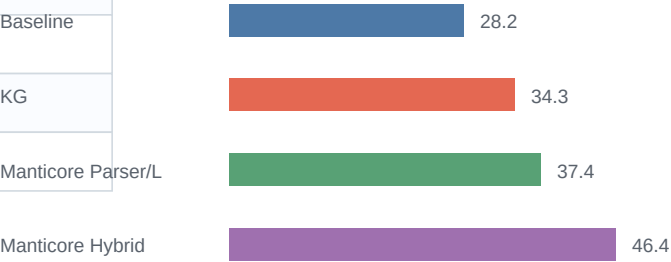
The Manticore rows use a local BM25F proxy and emit load-ready Manticore documents/schema.

Method	Input signals	Ranking	Strength	Tradeoff
Baseline	Module name, file path, ports, instances	Parser/LSP lexical overlap with exact anchors	Fast simple lookup, transparent evidence	Weak on semantic paraphrase and function similarity
KG	Baseline fields plus labels, summaries, graph context	KG-aware scorer with semantic expansion	Uses ontology and reverse graph relationships	Can add noise when labels are broad or sparse
Manticore Parser/LSP	Parser/LSP fields only	Manticore-style BM25F full-text field weighting	Best text ranking without KG dependency	Still blind to ontology-only relationships
Manticore Hybrid	Parser/LSP fields plus KG labels, summaries, parents	BM25F with KG field boosts	Best overall Hit@1 in this run	Slowest local proxy query latency

Hit@1 by Difficulty Level

Method	L1	L2	L3	L4	L5
Baseline	0.8571	1.0000	0.8571	0.6857	0.8571
KG	0.8286	1.0000	0.8571	0.7143	0.8571
Manticore Parser/LSP	1.0000	1.0000	0.8000	0.6571	0.8571
Manticore Hybrid	1.0000	1.0000	0.7714	0.7143	0.8571

Avg. Query Latency (ms)



Reading: Manticore Hybrid has the best Hit@1 and weighted Hit@1, while Manticore Parser/LSP has the best MRR. The hardest band is L4 across all methods; KG improves L4 Hit@1 over baseline, but hybrid ranking wins overall.